

UPI Fraud Trail Verification & Reconstruction

Investigator-facing graph analytics for UPI • NEFT • IMPS
mule-account trails

Et Tu Brutus!?

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DETECT

TRACE

EXPLAIN

REPORT

**A prototype that turns raw
transaction records into an
explainable money-flow investigation.**

The problem is already national-scale

₹22,495 Cr

reported due to cyber fraud

Amount reported by citizens on NCRP/CFCFRMS in calendar 2025.

24.02 lakh

financial-fraud complaints

Official 2025 NCRP complaint count; dynamic administrative data.

32.08 lakh

Layer-1 mule accounts shared

Shared by I4C with participating Suspect Registry entities by 30 Jun 2026.

The investigation bottleneck

A suspicious account is only the start. Investigators still need to reconstruct:

01 Where the money came from

02 Where it moved next

03 Which accounts form a ring

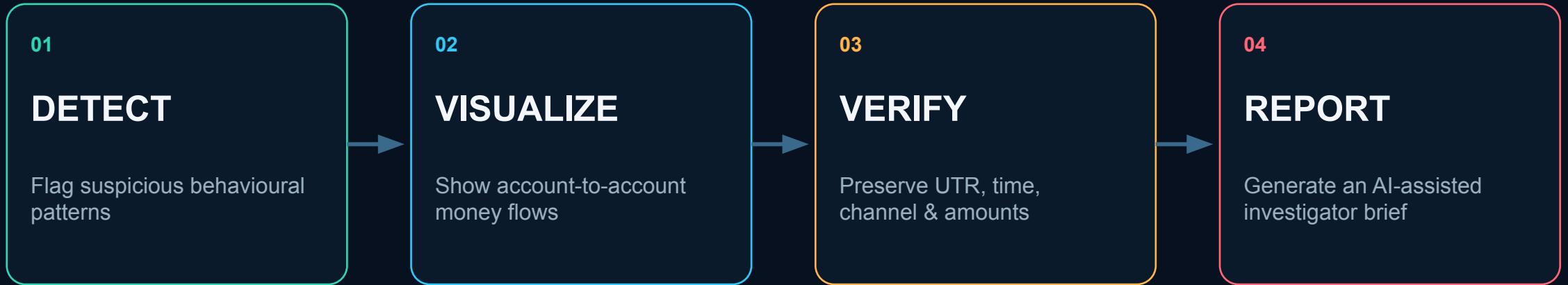
04 Why a flag is explainable

A flag is not proof of criminality — it is a prioritisation signal for human review.

Source: MHA / PIB — NCRP & CFCFRMS statistics (2025); I4C Suspect Registry (30 Jun 2026)

From transaction dump to investigation story

Our USP: investigator-facing reconstruction, not just account classification.

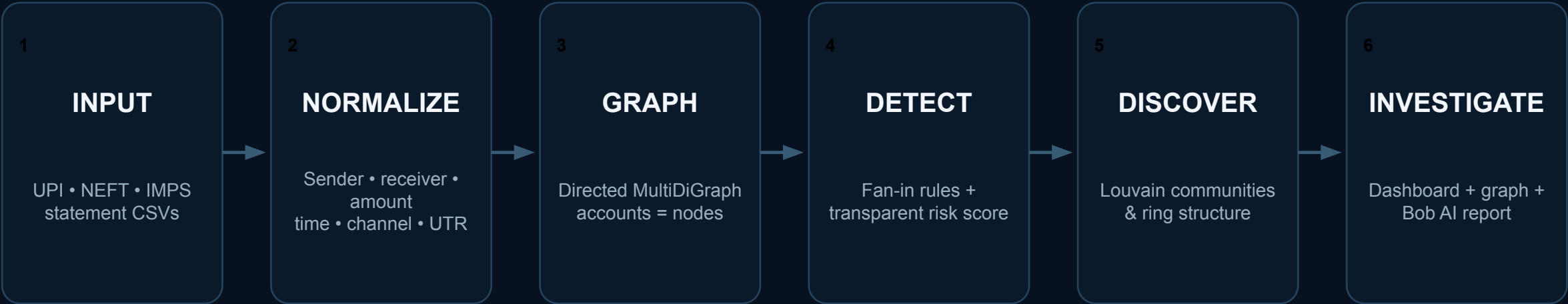


What makes it different Bank-side systems can identify risk. Our prototype focuses on the officer's next question: "Show me the trail, the connections, and the reason."

Complementary to systems like RBIH MuleHunter — whose published target segment is banks and bank fraud/AML infrastructure.

Source: RBIH MuleHunter documentation — target segment: banks; integration with EFRMS / AML systems

Architecture: transparent graph analytics pipeline

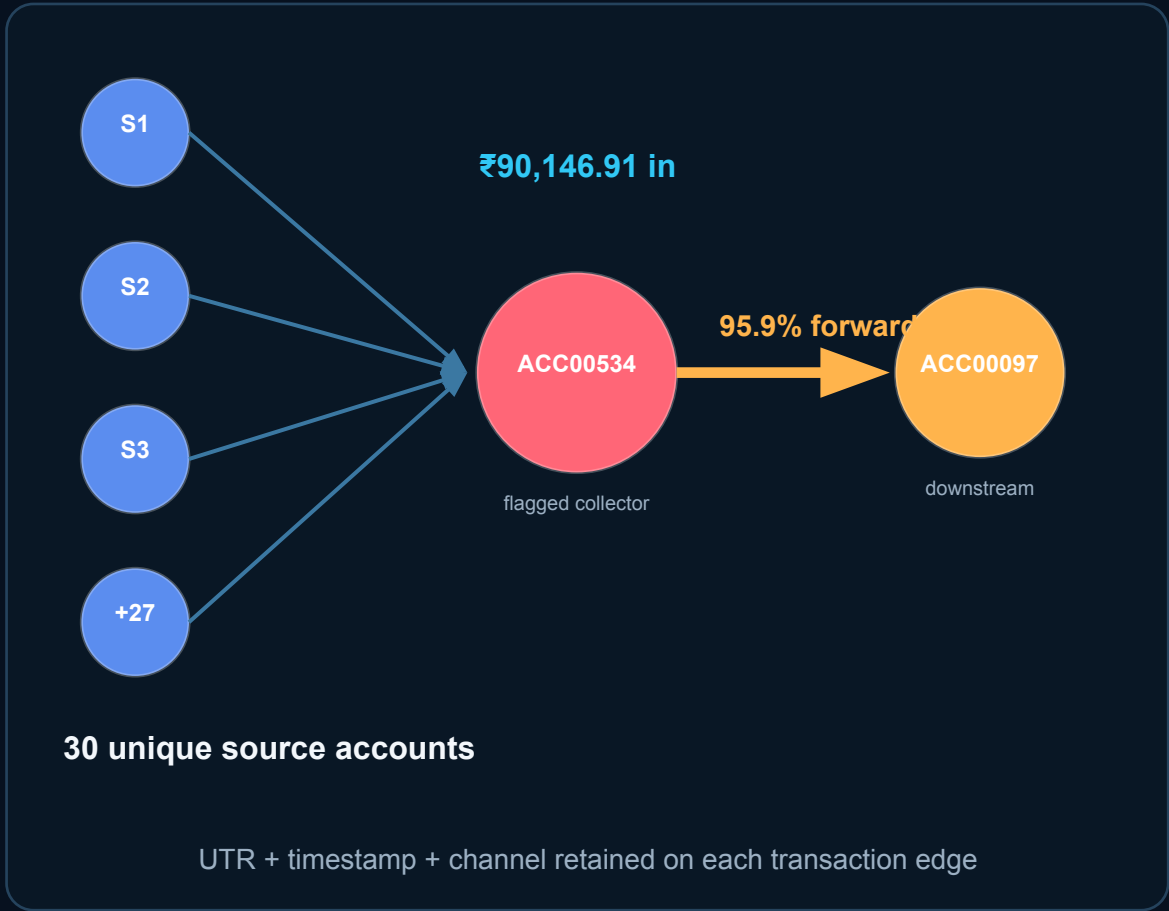


Why this is explainable	18 features	Stress-tested	2-feature score	UTR trail
	computed per account	camouflage activity injected	out-degree + unique receivers	preserved for verification

Prototype source: 20,479 normalized transactions; 548 statement files / graph nodes after ingestion.

One click: from suspicious node to explainable case

Synthetic demo case • ACC00534 • Fan-In Collector



Auto-generated investigator brief

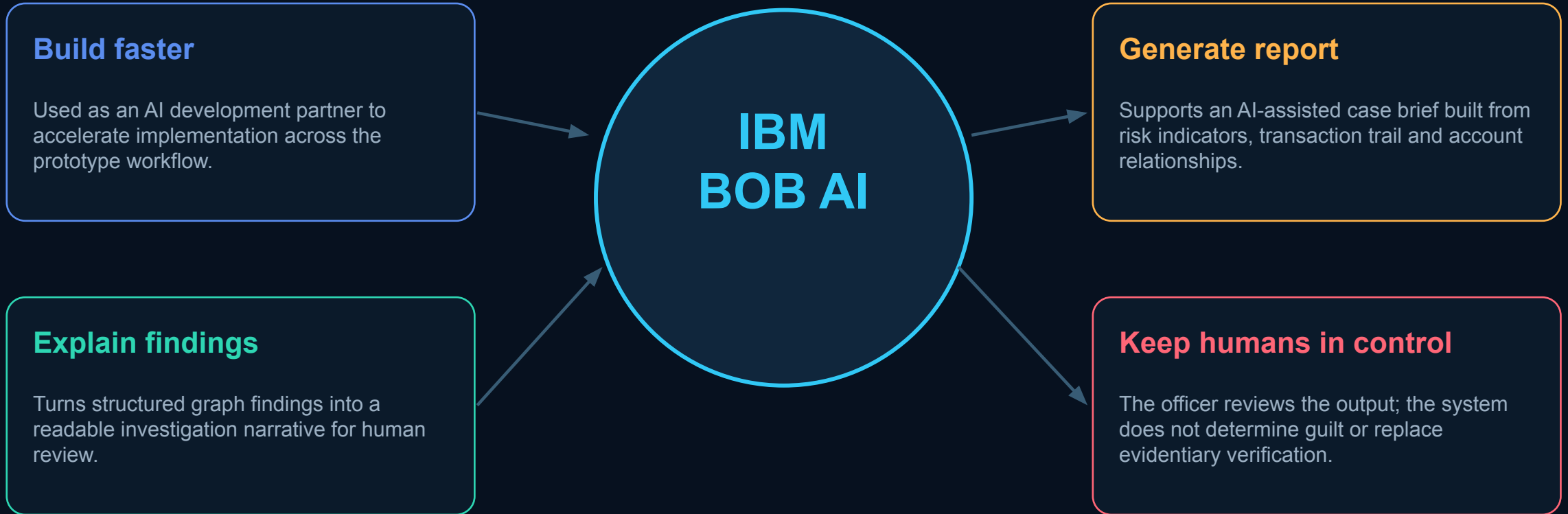
FIRST INFORMATION REPORT (FIR) - CASE BRIEF	
Cybercrime Financial Investigation & Evidence Summary	
Case Reference	CF/2026/534
Date of Brief	28 September 2026, 02:24 PM
Suspect Account	ACC00534
Detection Rule	Fan-In
Fraud Pattern	Fan-In Collector (Aggregation Fraud)
Activity Window	19 Jun 2026 – 19 Jun 2026
Risk Score	0.980 (CRITICAL)
Fraud Cluster	Community 1 (Louvain Unsupervised)
1. MODUS OPERANDI & NARRATIVE	
The suspect account ACC00534 operated as a fund aggregation node, receiving deposits from 30 unique source accounts totalling Rs 90,146.91. Approximately 96% of incoming funds (Rs 86,449.95) were rapidly forwarded to 1 downstream account(s), consistent with the modus operandi of a mule/collector account in a UPI/IMPS fraud syndicate. The high fan-in ratio (30 senders - 1 receiver) is a strong structural indicator of fund aggregation for onward laundering.	
2. FINANCIAL SUMMARY & FLOW RATIOS	
Metric Description	Quantified Value
Total Incoming Funds Deposited	Rs 90,146.91
Total Outgoing Funds Dispersed	Rs 86,449.95
Fund Pass-Through Velocity Ratio	95.9% forwarded
Inflow Transaction Count	30 transactions
Outflow Transaction Count	1 transactions
Unique Source Accounts (Potential Victims)	30 accounts
Unique Destination Accounts (Handlers/Mules)	1 accounts
3. NETWORK ENTITY HIERARCHY (Fund Flow Map)	
Source Accounts / Victims (30): ACC00010, ACC00020, ACC00065, ACC00071, ACC00078, ACC00085, ACC00097, ACC00115, ACC00123, ACC00136, ACC00142, ACC00144 ... and 18 more	
Mule / Aggregator Account: ACC00534	
Immediate Handler / Cashout Account(s) (1): ACC00097	

GRAPH → NARRATIVE → PDF

The report is an investigative aid — not an automatic finding of guilt.

IBM technology: Bob AI in the investigation workflow

The prototype uses IBM Bob AI as the IBM technology layer for development and AI-assisted report generation.



Source: [IBM Newsroom — IBM Bob: AI-first development partner across the SDLC](#)

Strong prototype signal — with honest limitations

95.8%

recall

46 of 48 synthetic mules caught

1.7%

false-positive rate

8 of 480 synthetic non-mules flagged

85.2%

precision

Fan-In + Tier-1 evaluation

7 / 7

fraud rings discovered

Louvain • zero labels used

What success looks like in deployment

- Prioritise accounts for analyst review
- Understand multi-account money trails faster
- Standardise a draft investigation brief

Validation status

Synthetic development data only
No independent holdout set
Small fraud sample
Real-world validation: pending

We present these metrics as prototype evidence — not production performance.

Et Tu Brutus!?

INNOVAI Hackathon by IBM

AK

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Detect → Visualize → Verify → Report